

Environmental drivers of low vaccine responsiveness in a lab-to-wild mouse model

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TODO LIST

	[SAB] Recompute NDE of $E := f(S)$ - see line 723	12
	[SAB] fit regression model of ACE and NDE of $P(E nP > 0)$	12
	[ABP] What do we plan to include in the supplemenatry methods?	13
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	Add details of the ISSF vaccination grant, Saudamini's Darwin Trust funding, ABPs Chancellors Fellowship, Amy Sweeny's Darwin Trust funding, SAB RS and BMGF. - what else?	20

ABSTRACT

Most vaccines fail to reach their intended target population, typically by failing late into the clinical trial pipeline. Even established vaccines may lose efficacy in certain individuals or groups that differ substantially from those in which they were initially tested, leading to breakthrough infections and the inability to achieve or maintain herd immunity. However, the causes of vaccine hyporesponsiveness remain difficult to identify, quantify, and therefore pre-empt. We hypothesise that the leading causes of vaccine hyporesponsiveness are environmental, and are exacerbated when individuals are required to allocate limited resources to growth, reproduction and immunity. Here, we show that circulating vaccine-specific antibodies dropped by 50% in a wild population relative to individuals of the same species that had been reared in laboratory conditions. Using paired cohorts of laboratory-reared and wild wood mice (*Apodemus sylvaticus*) vaccinated with a single and double-dose diphtheria toxoid vaccine (with adjuvant alum), we found that this loss in vaccine responsiveness was explained in the most part by diet, with weaker negative impacts of being male and reproductively active. However, helminth infection did not directly suppress vaccine-specific antibody production once adjusted for habitat and time since immunisation. While much variation remains to be explained, our results indicate that the environment plays a substantial role in vaccine failure, notably but not exclusively via nutrition, and that laboratory settings may systematically overestimate vaccine performance. We anticipate that our study inaugurates a robust method for disentangling and quantifying the complex factors that cause vaccine responsiveness in the wild. It may also call for a need to systematically raise efficacy thresholds for initiating clinical phases of vaccine development, and suggests that simple dietary interventions might improve individual responsiveness to immunisation.

INTRODUCTION

Unanticipated variability in individuals' responses to vaccination can lead to a lack of protection from the target pathogen, continued disease transmission, and an increased risk of vaccine escape¹⁻⁴. This heterogeneity is largely driven by the individual's pre-existing immune state, itself the outcome of complex interactions between intrinsic and extrinsic factors. Indeed, vaccine efficacy can be affected by genetics⁵, sex⁶, reproductive status⁷, past and current infections⁸⁻¹⁰, and nutrition¹¹⁻¹³. These factors can combine to affect both adaptive and innate immunity as demonstrated by the association between innate inflammatory markers in patients' peripheral blood before vaccination and subsequent antibody responses to 13 different vaccines¹⁴, and thereby, drive disease and transmission¹⁵. However, it is unclear how these factors interact and how to mitigate their effects to maximise vaccination efficacy. The limitations to our understanding of variability in vaccine efficacy stem in part from the inadequacy of current pre-clinical models used in vaccine development, which typically utilise defined inbred, non-reproductive, single sex, age-controlled animals that are bred in captivity under strictly controlled pathogen-free conditions. As a result, only a small minority of vaccines successfully transition from pre-clinical stages of development through phase III trials and reach mass market¹⁶ – in short, a generalisability and a transportability problem. Moreover, even successful vaccines can suffer from substantial variability between individuals and groups, notably for people of different ancestry¹⁷ or different habitats. For example, individuals of Tanzanian ancestry were reported to have higher inflammatory and type 1 immune markers¹⁸ and responses to measles vaccination¹⁹ than those of European ancestry, while in rural populations the expression of BCG- and tetanus-specific cytokine recall and antibody responses was found to be markedly lower than in urban populations, while humoral responses to tetanus vaccination were higher^{20,21}. Taken together, these reports indicate that vaccine efficacy depends on both the vaccine and the individuals recipient, and thus a need to develop

flexible and adaptable models that can be used to predict vaccine efficacy in different populations.

To increase the generalisability of experimental models of immunology and vaccinology, recent efforts have been made to integrate natural variability into immunological research^{22–26}. However, there is still uncertainty about how this heterogeneity affects the innate and adaptive arms of the immune system²⁷, and what the consequences might be for vaccine responsiveness. More specifically, it is unclear how environmental factors and past stressors influence organisms' responses to immunisation. This complexity blurs the causal relationships between environmental and immunological processes, making the design and interpretation of *in vivo* vaccination experiments difficult and their impact on their intended target populations intractable.

Here we propose a structural causal model (SCM)²⁸ to predict the effects of environmental variation on vaccine responsiveness and to identify the paths through which environmental and interventional effects on vaccine responsiveness are mediated. This causal framework, which utilises graphical models to inform adjustment strategies^{29,30}, can eliminate or reduce multiple sources of confounding, such as common causes (misattribution of causal effects), sampling biases (non-representative test population), and transportability biases (lack of generalisability). We performed an immunisation study using a coupled laboratory-wild wood mouse (*Apodemus sylvaticus*) system, which allows longitudinal sampling as well as experimental manipulation in both controlled laboratory conditions and in their natural woodland habitat. This allowed us to use the same assays and interventions across habitats and populations, and to replicate and condense the lab-to-clinic vaccine test-translate pipeline. We then compared the generation of vaccine-specific antibody responses to an alum-adjuvanted diphtheria toxoid vaccine (DTV) between laboratory and natural settings and between two diets (supplemented and control) after vaccination. We have previously demonstrated that this high quality, whole diet supple-

mentation drives a sustained reduction in helminth (*Heligmosomoides polygyrus*) burdens and transmission, while also improving the efficacy of anthelmintic treatment³¹. We therefore predicted that vaccine responses in wild mice would be lower than in lab wood mice, higher in diet-supplemented animals, and thus that diet might “rescue” hyporesponsiveness in wild rodents.

To identify and quantify the drivers of vaccine responsiveness in laboratory and wild settings, we used Bayesian SCMs to quantify the physiological and environmental direct and indirect causal effects of variation in the ability of recipients to generate vaccine-specific antibody responses (vaccine responsiveness). This allowed us to identify the main causes of loss of vaccine efficacy when translating a vaccine from the laboratory to the wild, and to accurately model how uncertainty and biological variability propagate along the causal network (see casual flow diagram, Fig. S1).

RESULTS

Vaccination induced stronger but variable antibody responses in the laboratory than in the wild

To assess how a laboratory environment, enriched diet, and vaccination protocol affect vaccine responsiveness, we designed a paired cross-factorial study in wild and laboratory *A. sylvaticus* in which mice were provided access to high or low-quality diet, and then immunised with a diphtheria-alum vaccine (DTV) in a single or double shot (see details in Materials, Methods, and Models; Figure 1).

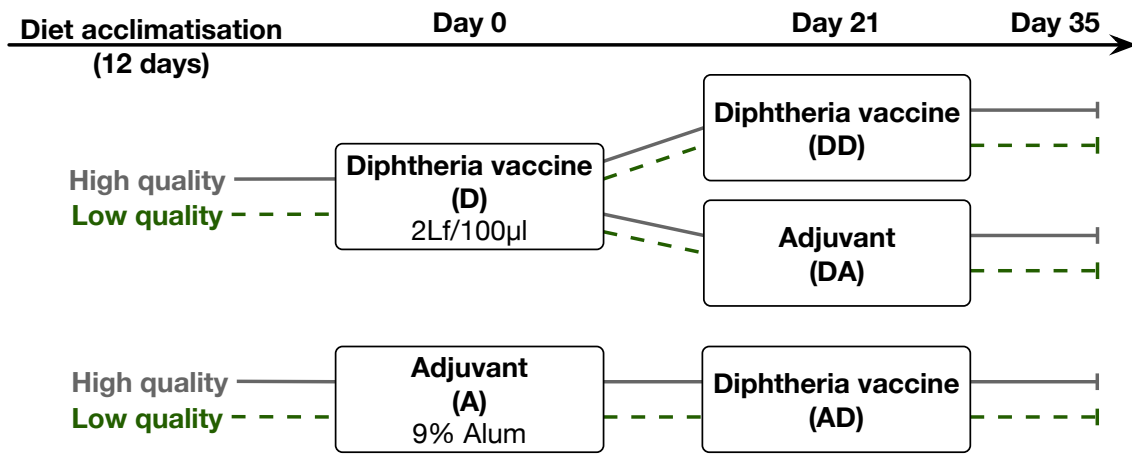


Fig. 1: Allocation of diet and vaccines. Wood mice from a laboratory colony, and from the wild were provided enriched (TransBreed) or standard (lab: normal chow; wild: no supplementation) diets. After at least 12 days, mice were allocated at random to immunisation with a 2Lf/100µl Diphtheria Toxoid + 9% alum vaccine (D) or with adjuvant alone (A). Twenty-one days after their first immunisation, mice that had been immunised with D were then given a second shot of the vaccine (DD) or adjuvant alone (DA), while mice that had received an initial adjuvant vaccine (A) were all given D for their second shot (AD).

Wild mice generated $46.9\% \pm 0.1\%$ lower DTV-specific IgG1 antibody responses than their lab-based conspecifics, irrespective of vaccine regimen or diet (Figure 2A, 2B ‘Hab’). While immunisation was most effective with two doses in both habitats, lab mice reach peak antibody levels with only a single dose (Figure 2A, ‘DA’ vs ‘DD’; Lab boosted vs. Lab non-boosted $OD = 0.2 \pm 0.1$ whereas Wild boosted vs. Wild non-boosted $OD = 1.02 \pm 0.23$). However, in the wild, only when given two doses did the mice show a level of responsiveness similar to that observed in the laboratory mice (Figure 2A, ‘DD’; difference between Wild non-boosted vs. Lab non-boosted $OD = 0.75 \pm 0.11$ while difference between Wild boosted vs. Lab boosted $OD = 0.61 \pm 0.18$; LRT of Habitat $\times$ vaccine regimen: $\Delta dof = 4, \chi^2 = 30.6, p < 0.001$). Only mice that were administered the DT antigen generated DTV-specific IgG1, indicating that there was no pre-existing immunity to this antigen in our study populations (Figure 2A, group ‘A’).

To further quantify how habitat (lab vs. wild), diet (supplemented vs. control), and immunisation protocol were associated with variation in vaccine responsiveness to each of the vaccine formulations, we constructed a multilevel Bayesian model of those factors

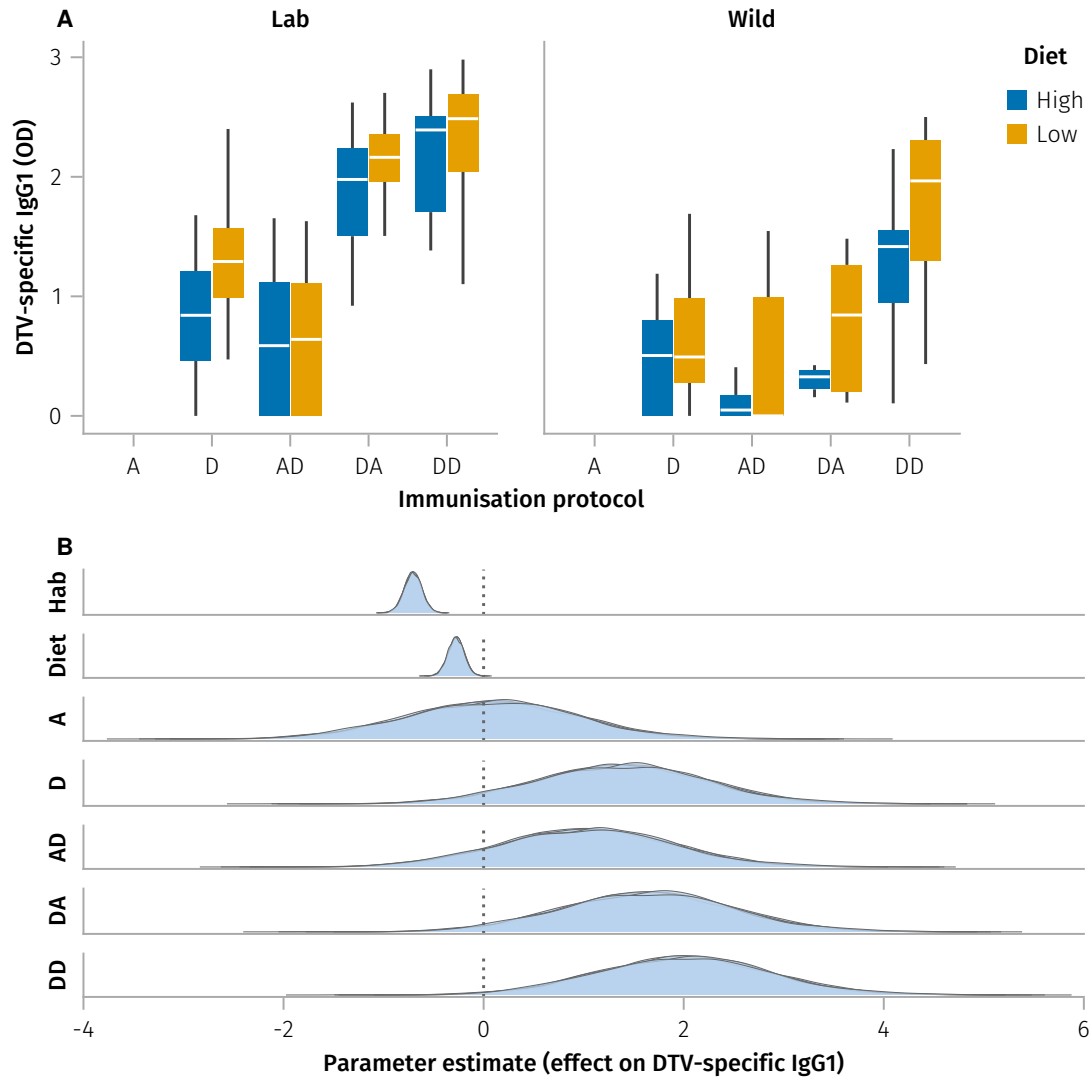


Fig. 2: Vaccine-induced DTV-specific IgG1. **A**, diphtheria toxoid vaccine (DTV)-specific IgG1 optical density in mice from laboratory- and wild-habitats offered high and low quality dietary supplements. Mice were immunised either once, with adjuvant as control (A) or DTV (D) only, or twice (i.e. with booster), using adjuvant alone followed by DTV at subsequent recapture ≥ 7 days after the previous immunisation (AD), DTV and subsequently adjuvant alone (DA), or two separate doses of DTV (DD). Only mice immunised at least 7 days prior to antibody measurement are shown, and each point represents one of 1–4 recaptures for each individual. **B**, distribution of posterior distributions of the coefficients for diet, habitat, and vaccine formulation (A, D, AD, DA, DD) of a multilevel Bayesian model adjusted for repeated individual measures. Each density curve represents the distribution of 3,000 samples from one of four MCMC chains (see Fig S2).

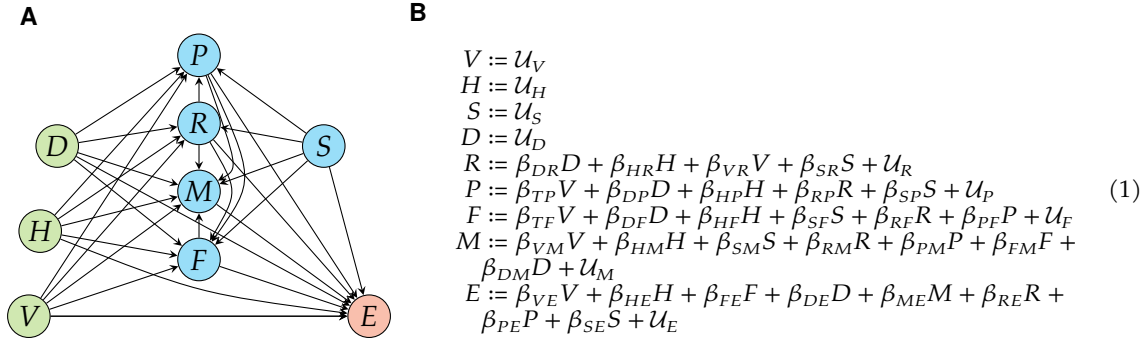


Fig. 3: Graphical Causal Model of vaccine efficacy. **A**, Directed acyclic graph representing hypothesised causal effects expressed by the SCM $\mathcal{C}$ (Model 1, right) driving variation in efficacy E of vaccine V in laboratory or wild mice H , under supplemented or control diet D (green vertices depict those that were experimentally intervened upon). Covariates and potential confounders (blue vertices) were time since vaccination T , parasite infection (presence or burden) P , reproductive status R , body mass M , fat scores F , and sex S . Directed edges depict causal effects hypothesised to operate between vertices. **B**, Assignments derived from the DAG with coefficients β_{ij} representing the effect of cause i on effect j . $\mathcal{U}_j$ represents the unmeasured variation of the response variable, not represented in the DAG panel A for clarity.

with intercepts for each immunisation protocol (Figure 2B, Table 1). This indicated that the effects of each vaccine formulation on DTV-specific IgG1 OD were substantially more variable than those of habitat and diet. However, this model, while it describes associations between habitat, sex, diet, and vaccine responsiveness, it fails to describe the biological processes that generate the outcomes of interest. Further, it does not account for potential confounding effects of sex, reproductive status, body mass, body fat, or parasite burdens, which we hypothesised to contribute to variation in vaccine responsiveness. Only an explicit causal model can help account for these effects and identify the main drivers of vaccine responsiveness.

Construction of a structural causal model of vaccine responsiveness

Next, to understand more specifically the relative contributions of diet and habitat to vaccine responsiveness, and how that effect might be mediated by parasite infection, reproductive status, and body condition, we built a causally-explicit model of the hypothesised causal relationships between (Figure 3).

To estimate vaccine responsiveness E as a function of vaccination V , habitat H and diet supplementation D , and more importantly, to identify which variables might act as con-

founders in need of adjustment or mediators that might modulate vaccine responsiveness, we constructed a directed acyclic graph (DAG, Figure 3A) to represent the hypothesised structure of the biological processes that generate the data observed in this system. We used this DAG to specify the structural causal model $\mathfrak{C}$ (Figure 3B) that links our variables of interest to vaccine efficacy E and describes the flow of causal effects based on the following criteria. Vaccination V being allocated at random, has no parent node. Its effects were hypothesised to be affected by habitat H since date of capture in the wild was inherently more variable than date of procedure in the lab for logistical reasons. A mouse being born in the lab or in the wild habitat H was considered random, and therefore took no causal input but was hypothesised to potentially affect all other variables except V and sex S . Diet D was allocated at random in the laboratory colony, but because wild mice could also forage freely, we considered that being born in the wild could have a causal effect on diet $H \rightarrow D$, specifically on the effect of the supplementation relative to the control laboratory or wild diet. Diet was hypothesised to affect reproductive status R , body mass M , body fat F and parasite burdens P as well as E , with fat scores F hypothesised to be a cause of variation in body mass M . Sex S being genetically determined and therefore subject to Mendelian randomisation³², and all treatments being randomised with respect to sex, S was considered causally prior to reproductive status R , body mass M , body fat F and parasite burdens P . Reproductive status R being driven by sex, it was considered causally prior to body mass M and parasite burdens P . Vaccine efficacy E was hypothesised to be potentially affected by all other variables. Each node also had its own error term to capture all unobserved sources of variation, assumed to be i.i.d To estimate vaccine responsiveness E as a function of vaccination V , habitat H and diet supplementation D , and more importantly, to identify which variables might act as confounders in need of adjustment or mediators that might modulate vaccine responsiveness, we constructed a directed acyclic graph (DAG, Figure 3A) to represent the hypothesised structure of the

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We then tested the validity of the causal hypotheses described above using the DAG to construct each model. For example, the DAG in Fig. 3 implies that parasite infection should be conditionally independent of vaccination if we adjust for diet, reproductive status, sex, and habitat ($P \perp\!\!\!\perp V \mid D, R, S, H$); we therefore fit a binomial generalised linear model $glm(P \sim \alpha + \beta_V V + \beta_D D + \beta_R R + \beta_S S + \beta_H H)$. This process was repeated for all implied independencies derived from the causal graph (see Model validation methods),

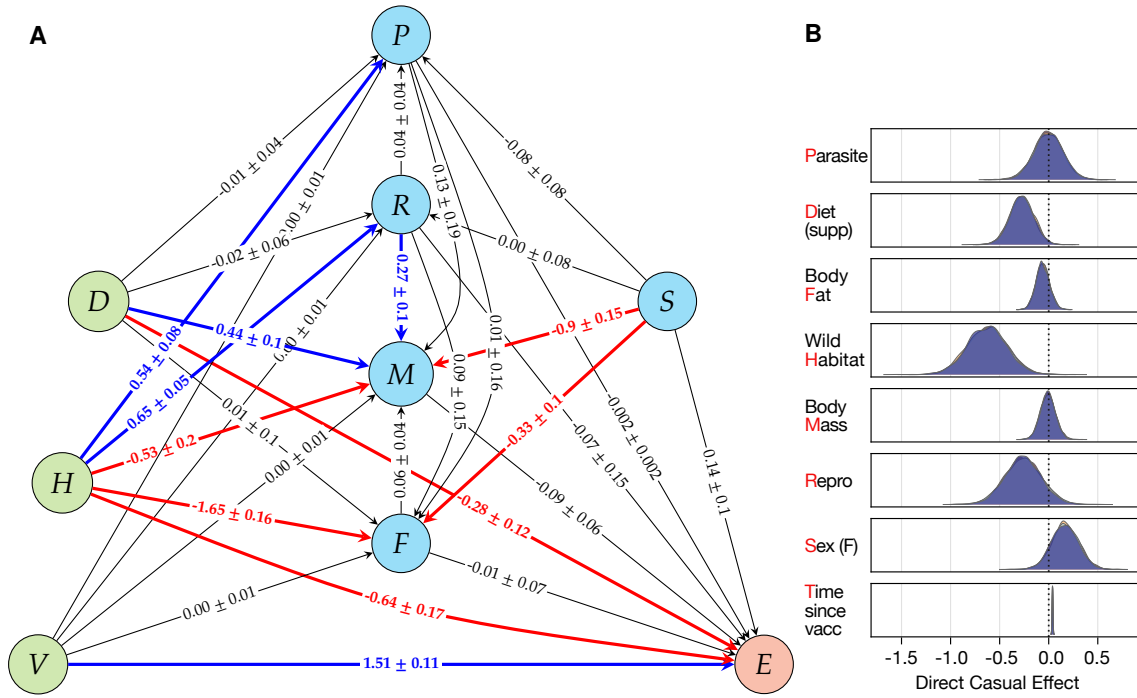


Fig. 4: Fully resolved Causal Model of vaccine efficacy. **A**, Directed acyclic graph representing point estimates of positive (blue) and negative (red) direct causal effects $\pm$ std driving vaccine responsiveness E . **B**, MCMC sampling of posterior distributions of the coefficients for $E := \beta_{DE}D + \beta_{FE}F + \beta_{HE}H + \beta_{ME}M + \beta_{PE}P + \beta_{RE}R + \beta_{SE}S + \beta_{TE}T$, the linear estimation of $P(E|Z)$, where Z is the adjustment set (see Model parameterisation methods). Note that each edge in panel A shows the direct causal effect of its parent node on its child node, which is conditional on the specific adjustment set for that model; thus, only the coefficient for the edge between D and E is a causal estimate in panel B, while the others are only associative.

and the DAG updated until all causal paths were supported by the observed data and consistent with plausible causal expectations. This resulted in the DAG shown in Figure 3.

Average, direct, and indirect causal effects of immunisation on vaccine responsiveness

We fully parametrised $\mathcal{C}$ (Model 1) using a set of Bayesian linear models containing variables required to ensure de-confounded estimation of the total and natural direct causal effects operating between the nodes of $\mathcal{C}$ (Fig. 4). Vaccination resulted in an average causal effect of 1.51 ± 0.11 ($\log_{10} + 1$)-transformed OD units ($\log OD$) from unvaccinated animals in this experiment, 1.49 ± 0.11 of which were estimated as direct (or mediated by unobserved variables, e.g. immune pathways) with a negligible fraction (<0.1 on average) mediated by reproductive status R , body mass M , and body fat F (Fig. 4B). However,

vaccine responsiveness E was strongly influenced by both the wild habitat and dietary supplementation. The wild habitat had a direct negative impact of $0.64 \pm 0.17 \log OD$ on vaccine responsiveness, while dietary supplementation resulted in an average reduction of $0.28 \pm 0.12 \log OD$.

Furthermore, while wild wood mice were lighter (-1.3 ± 0.2 g on average) and accumulated less fat (by 2.3 points on 6-point scale), dietary supplementation increased body mass ($+2.53 \pm 0.3$ g) though did not affect body fat scores. Interestingly, the effect of dietary supplementation on vaccine responsiveness E was mostly direct (or mediated by unobserved variables) and not mediated by its effects on body mass, fat stores, reproductive status R , or parasite burden P . The total causal effect of sex on E indicated greater vaccine responsiveness in females ($0.26 \pm 0.15 \log OD$).

[SAB] Recompute NDE of $E := \hat{f}(S)$ - see line 723

Effect of parasite infection among the infected

When considering the whole population including uninfected mice, *H. polygyrus* burden had no clear effect on DT vaccine responsiveness despite being strongly positively associated with the wild habitat (Fig. 4B). However, because parasite infections are overdispersed or even zero-inflated, we split the analysis into estimating both the causal effects of infection burden on DT vaccine responsiveness among the infected.

[SAB] fit regression model of ACE and NDE of $P(E|nP > 0)$

DISCUSSION

The physiological variables measured mediated very little of the effect of immunisation on vaccine responsiveness, if any. However, environmental covariates of supplemented diet and wild habitat had substantial negative effects on vaccine responsiveness.

The 'wild habitat' captures many variables that would still need to be identified - these might include weather, predation, competition, and other stressors that are absent from laboratory conditions.

Effect of nutrition was unexpected. See¹³

Weak / variable effect of worms^{9,10,33}.

Limitations & Perspectives. We made the simplifying assumptions that:

- parametric assumptions and linearity of the causal effects. Future work will need to generalise the functional form of the causal effects to allow for non-linear effects of the variables on vaccine responsiveness.
- no unmeasured confounders. Future work will need to relax these assumptions to allow for more complex causal structures and to account for unmeasured confounders and mediation.
- random vaccine and diet treatments: we should allow for vaccine and diet uptake (propensity scores) to vary among the treated, as especially in the wild population, these are not strictly enforced.

MATERIALS, METHODS, AND MODELS

All animal work was conducted in accordance to UK Home Office in compliance with the Animals (Scientific Procedures) Act, 1986. Both laboratory and field experiments were approved by the University of Edinburgh Ethical Review Committee and carried out under the UK Home Office Project Licence 70/8543. The dosage of diphtheria vaccine was trialled in laboratory reared wood-mice before the experiments for safety and efficacy. Animal sacrifice was performed using appropriate Schedule 1 Methods. Fieldwork was carried out with permission of the Forestry Commission Scotland under the permit SUR09.

See Supp methods

[ABP] What do we plan to include in the supplementary methods?

Laboratory wood mouse experiment

The laboratory experiment was conducted on a wild-derived but now captive, outbred wood mice colony that has been bred and maintained at the University of Edinburgh³¹ (See SI for more details). The experiment was carried out in two replicate blocks, with 36

animals in each block (18 males, 18 females). At 12 days before the start of the experiment (d-12), all mice were shifted to new diet regimes and given time to acclimatise; half of the animals in each block (9 males and 9 females) were given TransBreed™, a high-quality whole-diet mouse chow. We have previously shown that wood mice supplemented with TransBreed™ were more resistant to the gastrointestinal nematode *Heligmosomoides polygyrus*, cleared worms more effectively after anthelmintic treatment and produced stronger general (total IgA) and parasite-specific (IgG) immune responses in both wild and laboratory conditions³¹. The other half of the mice in this experiment were fed RM1 (Rat and Mouse Maintenance), a lower quality diet, but commonly used maintenance mouse chow which supports maintenance. Both TransBreed™ and RM1 are standard laboratory diets, with TransBreed™ containing higher amounts of starch, protein, fat and micro-nutrients compared to RM1 (see SI for detailed nutritional compositions). Within each diet regime, both male and female animals were randomly assigned to three experimental groups (Fig. 1). Animals belonging to Groups 1 & 2 (n = 24) were subcutaneously injected with 100µl of diphtheria vaccine (2 Lf per 100µl of adjuvant (9% Potassium Aluminium Sulphate) on d0. The vaccine was prepared in our laboratory one day before administration, using vaccine-grade diphtheria inactivated toxoid protein (Alpha Diagnostic International) adsorbed to the adjuvant (9% Alum). Animals assigned to Group 3 (n = 12) were injected with a control, referred to as 'adjuvant'; 100µl of adjuvant (9% Potassium Aluminium Sulphate) suspended in 1X PBS on d0. Small volume blood samples were taken via tail snip 14 and 17 days after the first injections (d14 and d17) to measure the primary antibody response. Twenty-one days (d21) after the first injections, animals in Group 1 (n = 12) were injected again with 100µl of the diphtheria vaccine (referred to as the 'booster'), whereas those assigned to Group 2 (n = 12) were injected with the control dose as described above (referred to as 'adjuvant'). Animals assigned to Group 3 (n = 12) that had previously been injected with the control dose, now received an injection of 100µl diphtheria vaccine.

Another blood sample (20- 50µl) was taken a day later (d22) via cheek venipuncture. All animals were sacrificed 14 days after the second injection (d35) and a blood sample for measuring the secondary antibody response was taken. The same procedures and timeline were carried out for the second experimental block of animals (n = 36). Mice were co-housed throughout the experiment, in same sex groups of three, with equal representation of the three experimental groups in each cage.

Wild wood mice experiment

We conducted a 10-week field experiment in a natural population of wood mice with a design similar to that of the laboratory experiment. The experiment took place in a woodland in Falkirk, Scotland, UK (Callendar Wood, 55.990470, -3.766636), where we established four trapping grids (60 m X 40 m per grid) with 10m spacing between each of the 35 trapping stations. At each station, we set a pair of Sherman live traps (H.B. Sherman 2X2.5X6.5-inch folding trap, Tallahassee, FL, USA). We randomly selected two of the four grids to be ‘supplemented’; 12 days before we began live-trapping, we started supplementing these grids with extra resources. Specifically, we scattered 6 kg TransBreed™ (high quality, whole- diet mouse food) pellets evenly around the grid, and this supplementation continued throughout the experiment, twice per week. The two ‘control’ grids did not receive any supplementation. Wood mice on all four grids had access to their normal diet and food items, so this extra resources on the ‘supplemented’ grids served as an additional high nutritional food resource.

From July-September 2018, wood mice were trapped 2–3 nights per week using live traps baited with grains, carrots, and bedding. Additionally, traps on the supplemented grids were also baited with 1-2 TransBreed™ pellets. For identification, all newly captured mice weighing >13g were subcutaneously tagged with a unique 9-digit microchip passive induced transponder (PIT tag; FriendChip AVID2028, Norco, CA, USA). On first capture, mice were assigned to one of the three vaccine treatment groups described above in the

[SV] Could refer to ARS's work that shows reduced helminth burdens in diet-supplemented wild mice

[ABP] Agreed. I added a few more details above - but let me know if you think we need to add more

laboratory experiment; mice assigned to these groups received the same primary and booster doses of the diphtheria vaccine or control injections as described below (see Figure 1). Throughout the experiment, weekly small volume blood samples were taken via tail snip from each tagged animal to measure their antibody response to the diphtheria immunisation. Animals recaptured 14 or more days after their second injection were sacrificed and a terminal bleed was collected to measure their antibody response and adult *Heligmosomoides polygyrus* worm burdens. *H. polygyrus* is a natural gastrointestinal nematode found at high prevalence (20-100 %) in wild populations of *A. sylvaticus*^{31,34–36}, as well as being a well-studied model system of human gastrointestinal nematodes where it has been found to be highly immunomodulatory^{37,38}.

Additionally, at all captures, we took the following demographic metrics for each individual: sex, body condition, reproductive condition, weight (g), and length (mm). Sex and reproductive condition of each mouse was assigned by examining the genitals, with males having larger urogenital gap compared to females. Males were classified as reproductively active if their testes had visibly descended to scrotal sacs, and females were classified as reproductively active if they had a perforated vagina or were visibly pregnant or lactating. Body condition was measured by assigning dorsal and pelvic fat scores on a qualitative scale from 1-5, where 1 represented a very low condition mouse, while a 5 represented a mouse with ample fat reserves. This was accrued out by palpating the back and pubic bones³⁹.

Vaccine responsiveness

Vaccine responsiveness was measured as the concentration of anti-diphtheria IgG1 antibodies; this is the gold standard method used in vaccinology to measure the antibody-specific response to immunisation and is used specifically for the diphtheria toxoid vaccine used here⁴⁰. To measure diphtheria vaccine responsiveness, we collected blood samples from both the laboratory and the wild mice and these were centrifuged (@24,000 rpm for 10

min) within 4–6 hours of collection. The sera were separated from the pellets and stored at -80 °C until further analysis. We used ELISAs adapted for diphtheria-specific antibodies from Clerc et al. (2019a), to quantify Anti-diphtheria IgG1 antibodies in the sera samples. 96-well plates (Nunc™ MicroWell™) were coated overnight at 4 °C with diphtheria toxin (2µg/mL) diluted in carbonate buffer (50 µl per well). After washing the plates with TBS (10 X) and Tween80 thrice, 100 µl of TBS(1X)-4percent BSA was added per well and incubated at 37 °C for 2 hours to block non-specific binding sites. Plates were then tapped dry and 50µl of serum samples serially diluted (starting dilution 1:100) in TBS (1X)-4percent BSA buffer were added per well and left at 4 °C for binding overnight. The next day, 50µl HRP-conjugated anti-mouse IgG1 detection antibody (Southern BioTech) was added per well after washing the plates 4 times with TBS (10X) and Tween80 and tapping them dry. After incubation at 37 °C for 1 hour, the plates were washed again, first 4 times with TBS (10X) and Tween80 and then twice with dH2O. 50µl of TMB substrate solution was added per well and the enzymatic reaction was left to develop in the dark for 7 min. The reaction was stopped after 7 min using 50µl sulphuric acid (0.18 M) per well. Absorbance at 450 nm was recorded using an ELISA plate reader (Multiscan, Ascent Labsystems) immediately after. For each sample, blank-centred optical densities (OD) of three consecutive wells (of dilutions 1:3200, 1:6400, 1:12800) were averaged to obtain diphtheria-specific IgG antibody concentration.

Data processing and analyses

DT-specific antibody ELISA optical densities (E) were $1 + \log_{10}$ -transformed and, body weight (M), fat scores (F) and days post-vaccination ($T \geq 1$), were Z score-transformed to help the selection of sensible priors and to improve computational stability. All models were adjusted for time T between vaccination and blood collection to account for natural kinetics of the antibody response. Fat scores ranged from 2 to 5 and were derived from the sum of two five-point scales for pelvic and dorsal fat, where 1 is ‘bony’ to palpation and 5

is ‘fat’, and were treated as continuous for the analysis. All other variables were treated as binary.

All data processing was performed in Julia v1.9⁴¹ using packages CSV.jl v0.10⁴² and DataFrames.jl v1.3⁴³.

Generalised linear mixed models of the effects of experimental interventions on habitat, diet, and vaccine formulation

Generalised linear mixed models (GLMM) were used to estimate the effects of habitat, diet and vaccine formulation on vaccine-specific antibody concentrations measured as optical densities read-outs. Mouse ID was modelled as a random effect to account for multiple testing. The interaction between vaccine regimen and habitat was included in the full model to test whether mice in the wild would respond differently to vaccine timing and boosting than laboratory mice. Likelihood ratio tests (LRT) were used to test the contribution of the interaction terms to the model. We used Julia v1.9⁴¹ package MixedModels.jl v4.13⁴⁴ was used for GLMMs and LRTs.

Bayesian statistics: choice of priors and sampling of posterior estimates

We used Bayesian multilevel models to minimise the effects of imbalance on our analyses and to quantify the uncertainty of those estimates given the available data. Priors for intercepts and slopes were chosen to conform to the observed data, with equal means and $2.5 \times$ the observed variance (see Supplementary Data: Multilevel Models). Typically, intercepts were given Gaussian priors of mean $\mu = \text{mean}(y)$ and $2.5 \times \sigma$ those of the observed response variable y while population-level coefficients were given multivariate normal (Gaussian) priors of mean $\mu = 0$ and variance $\sigma = 2$. Population-level residuals were assumed to follow an exponential distribution with scale $\theta = 1/\text{std}(y)$. Hyperparameters for group level intercepts and coefficients were given Gaussian priors and their variances were given positive-truncated Cauchy priors with location $x_0 = 0$ and scale $\gamma = 2$. Pos-

terior estimates were sampled using the Hamiltonian Monte Carlo algorithm No U-Turn Sampler⁴⁵, typically across 4 threads simultaneously with $> 3,000$ draws. Turing.jl⁴⁶ was used for Bayesian modelling; Makie.jl v0.17⁴⁷ was used for plotting.

Structural Causal Models

The structural causal model (SCM) framework^{28,48} proposes explicit qualitative solutions to biases inherent in datasets and can guide the statistical estimation of the mechanisms by which data are generated in complex systems^{49,50}. Further, Bayesian statistical methods for estimating the causal effects indicated by the SCM allows flexibility of imbalanced small data and missing values⁵¹, and provides explicit estimation of uncertainty given assumptions made by the model⁵². Specifically, here we have made three important assumptions: vaccine, diet, and habitat treatments are random; there are no (statistically relevant) unmeasured confounders; and all relationships between variables are linear.

Model construction

Each of the variables X described above was included in the structural causal model $\mathfrak{C}$ with the following assignments of the form $X_j := f(X_{k \neq j}) + \mathcal{U}_j$ where the unmeasured noise variable $\mathcal{U}_j \sim \text{Normal}(0, 4)$, was assumed to be independent and identically distributed, and $f()$ was assumed to be linear (Supplementary Data: SCM construction):

A directed acyclic graph (DAG) expressing $\mathfrak{C}$ (Model 1) was used to build unconfounded models for the estimation of causal effects between each variable (Figure 3).

Model construction was performed in Julia v1.9⁴¹ using package Dagitty.jl and Turing.jl⁴⁶, and the DAG corresponding to $\mathfrak{C}$ was drawn in Tikz⁵³.

Model validation

To validate $\mathfrak{C}$ (Model 1), we tested the independencies it implies following chain, fork, and collider rules^{28,54}. Specifically, we used the observed values for each of the nodes of

the causal graph (Figure 3) to test the truth of the following 13 implied independences:
 $(D \perp\!\!\!\perp S)$, $(D \perp\!\!\!\perp T \mid H)$, $(D \perp\!\!\!\perp V)$, $(F \perp\!\!\!\perp V \mid T, H)$, $(H \perp\!\!\!\perp S)$, $(H \perp\!\!\!\perp V)$, $(M \perp\!\!\!\perp V \mid T, H)$,
 $(P \perp\!\!\!\perp T \mid D, R, S, H)$, $(P \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp V \mid D, R, S, H)$, $(R \perp\!\!\!\perp V \mid T, H)$, $(S \perp\!\!\!\perp T)$, and
 $(S \perp\!\!\!\perp V)$. For each of these tests, we constructed a generalised linear mixed model with the
appropriate link function, with mouse ID and immunisation protocol as random effects,
and considered the statement true if $P_X > 0.1$, where $X \subseteq \mathcal{C}$ (Supplementary Data: SCM
construction). We used MixedModels.jl v4.13⁴⁴ to test the conditional independences
implied by $\mathcal{C}$.

Model parameterisation

Next, we encoded $\mathcal{C}$ as a Bayesian network in Turing.jl⁴⁶ to represent each variable as a
probability distribution to be updated by the observed values. This allowed us to capture
conditional expected values for each variable and their uncertainty. Posterior estimates
were validated with MixedModels.jl⁴⁴.

Missing data. Twenty-two mice had no fat scores, resulting in 67 missing values in the
full repeated sampling dataset. To avoid having to remove those mice from the analysis,
we used Bayesian imputation to estimate plausible probability distributions of each of
the missing values in light of its parents (PA) in $\mathcal{C}$ (Model 1), where $F := f(PA_F) =$
 $f(T, H, D, P, S)$ (Figure 3).

$\hat{F}$ denotes the variable F that contains imputed values; E_i denotes the expected distri-
bution of E given $\hat{F}$.

Model updating and sampling. All Bayesian modelling was performed in Julia v1.9⁴¹ using
package Turing.jl v0.25⁴⁶.

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funding, SAB
RS and BMGF -
what else?

COMPETING INTERESTS

411 The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

412 Conceived and designed the experiments: ABP, SAB

413 Performed the experiments: SV, AS, ABP, SAB

414 Analysed the data: SV, SAB

415 Contributed reagents/materials/analysis tools: ABP, SAB

416 All the authors reviewed the manuscript.

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TABLES

Table 1: Posterior predictions of model parameters for random intercept Gaussian model $E \sim \text{Normal}(\alpha[ID] + H + D + V, \sigma)$.

Parameter	Mean	Std	5%	50%	95%
Intercept	-0.476	0.851	-1.878	-0.470	0.902
Diet	-0.273	0.079	-0.402	-0.274	-0.142
Habitat	-0.704	0.087	-0.850	-0.703	-0.560
A	0.105	0.838	-1.257	0.100	1.480
D	1.391	0.836	0.027	1.385	2.770
AD	1.046	0.838	-0.321	1.046	2.424
DA	1.655	0.839	0.288	1.652	3.037
DD	2.017	0.840	0.650	2.014	3.404

Environmental drivers of low vaccine responsiveness in a lab-to-wild mouse model

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SUPPLEMENTARY INFORMATION

Supp methods

Insert additional details about the lab mouse colony details of the diet make up of trans-breed and RM1

Supplementary Data: Multilevel Models

Supplementary Data: SCM construction

Supplementary Data: SCM specification

Supplementary Data: Wood mouse data

Supplementary Figures

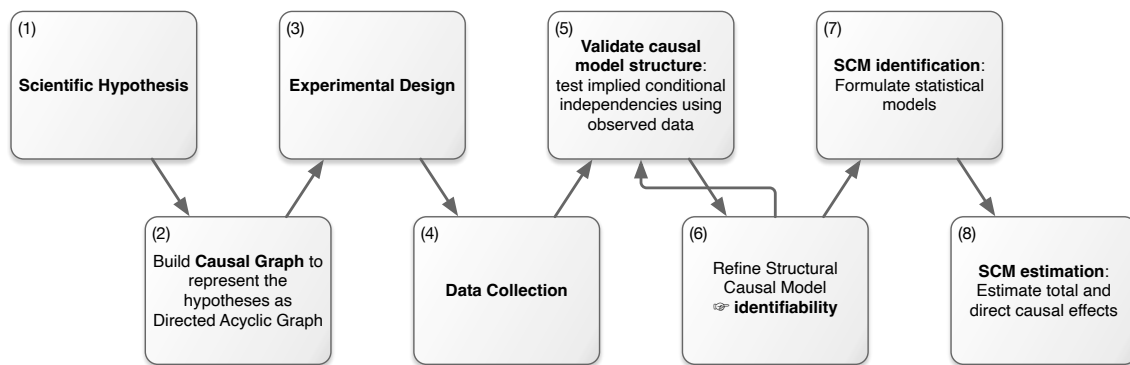


Fig. S1: Flow diagram of the SCM. The Structural Causal Model (SCM) process explicitly represents scientific hypotheses (1) as a directed acyclic graph (DAG). This DAG is mathematically encoded as a set of nested equations that describe the flow of causation between variables, and helps inform lab and field experimental design (3) and data collection (4). After processing, the data are used to test the validity of the causal assumptions underlying the SCM, e.g. by testing the conditional independencies implied by the DAG (5). If the assumptions are not supported, refinements of the SCM (6) are necessary. When all conditions are met, the SCM is identifiable and statistical models can be formulated to adjust for confounding (7). Parameters from these models can then be used as estimates of direct and indirect causal effects (8).

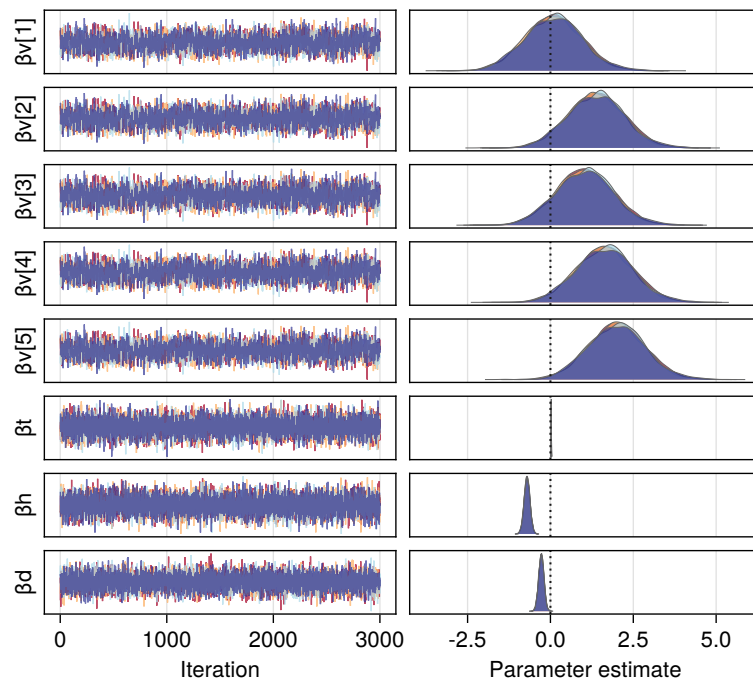


Fig. S2: MCMC chains and posterior distributions of coefficients for Diet, Habitat, Time post immunisation, and vaccine formulations A, AD, D, DA, and DD.